

♦ P2: Network Scanning and Reconnaissance

Aim

To perform network scanning and analyse network traffic.

Tools Required

Windows Command Prompt, Kali Linux (WSL), Nmap, Wireshark

Procedure

Step 1: Check Network Configuration

- Open Command Prompt in Windows and view network details.

```
C:\Users\rauna>ipconfig

Windows IP Configuration

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::c2f8:efc1:c95b:f28e%47
    IPv4 Address. . . . . : 172.22.32.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : hgu_lan
    Link-local IPv6 Address . . . . . : fe80::c2fd:c219:7f7b:868%4
    IPv4 Address. . . . . : 192.168.1.43
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

- Open Kali Linux and check network interfaces.

```
(kali@HP15s)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:c5:0c:40 brd ff:ff:ff:ff:ff:ff
    inet 172.22.34.37/20 brd 172.22.47.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fec5:c40/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Step 2: Test Connectivity

- Use ping command to verify internet connectivity.

```
(kali@HP15s)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=8.07 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=4.84 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=5.31 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=5.50 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=5.20 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=118 time=4.95 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=118 time=5.55 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=118 time=5.28 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=118 time=4.98 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=118 time=6.19 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=118 time=5.50 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=118 time=4.33 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=118 time=5.19 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=118 time=5.69 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=118 time=5.14 ms
```

Step 3: Perform Network Scanning

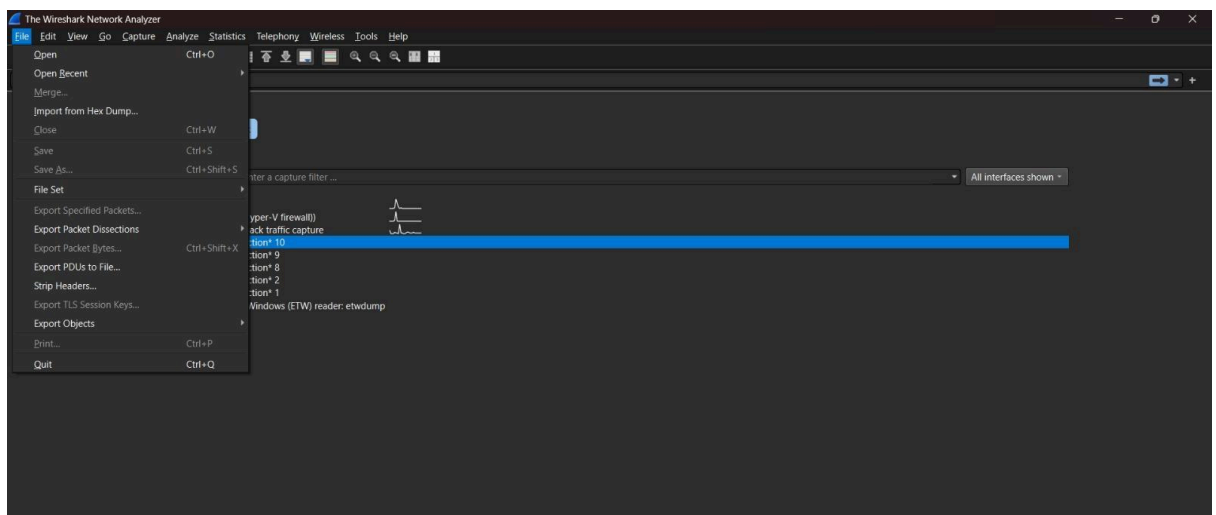
- Install Nmap in Kali Linux.
- Scan localhost for open ports.
- Perform full port scan.

```
(kali@kali:~)$ nmap 127.0.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-29 09:33 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
3000/tcp  open  ppp

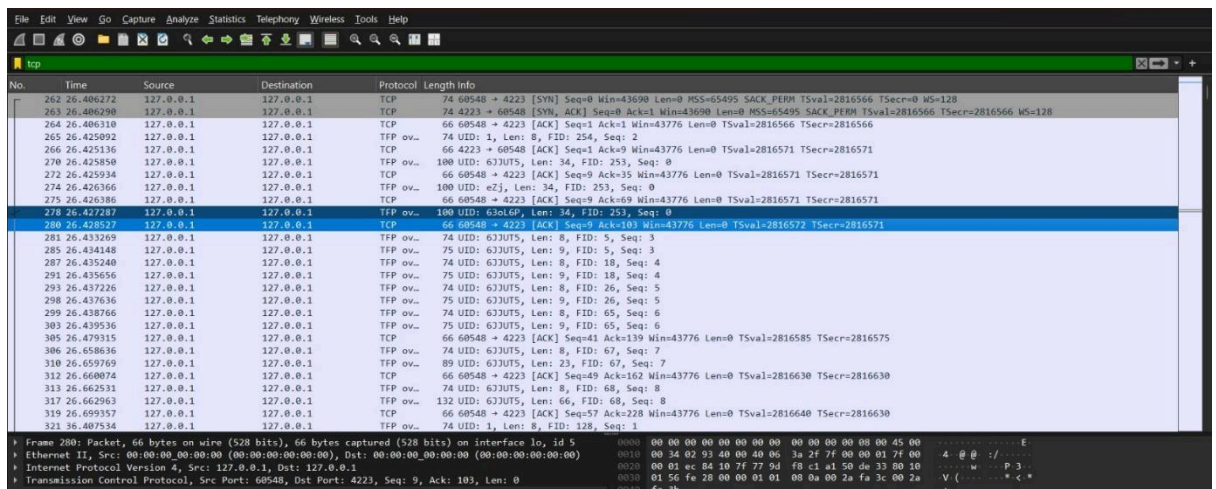
Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
```

Step 4: Packet Analysis using Wireshark

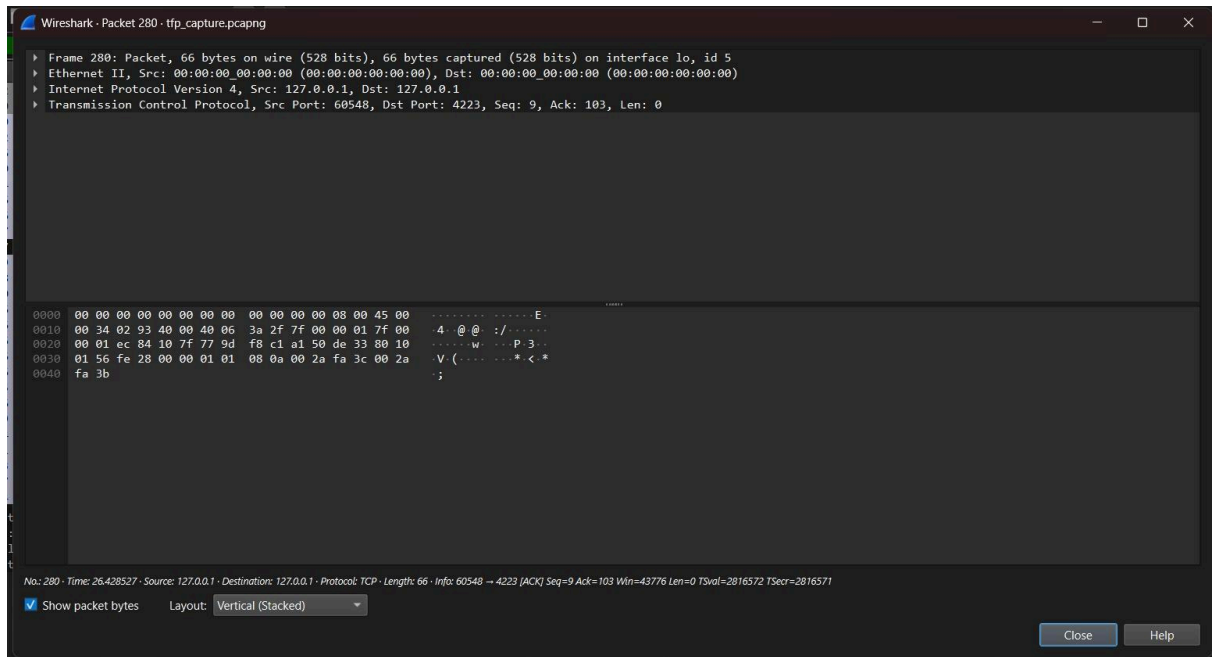
- Download a sample packet [capture file](#).
- Open it in Wireshark.



- Apply filters such as HTTP, TCP, DNS.



- Analyse packet details.



Result

Successfully analysed network configuration, scanned ports, and examined network packets.